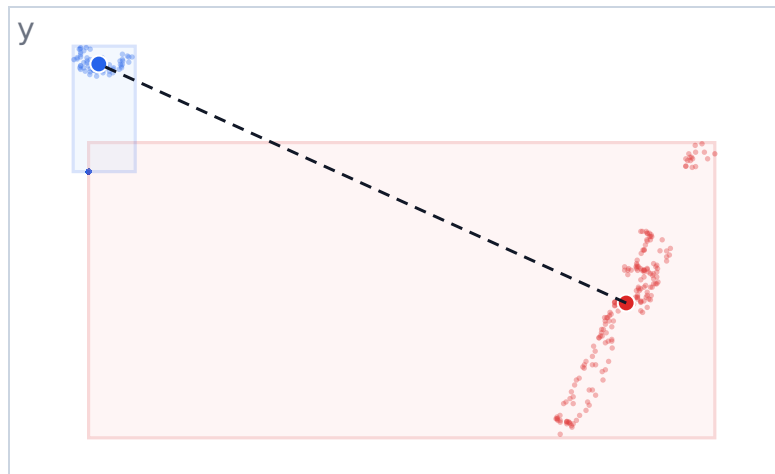


(a) High-confidence relation

● subject ● object



heater → close by → trash can

Open3DSG: $Z = 0.853$; rank 19

XY distance = 4.33 m

(b) Source-score-excluded compatibility

Predicate T

close by / proximity

Pair geometry G

distance, overlap, contact

Compatibility $C(T,G)$

linked counterfactual ordering
relation-algebra projection

$C = 0.003$

Z is not an input to C

Source score Z

used only in re-ranking

$S = Z \times C(T,G)$

within applicable family

(c) Re-ranking

19

source rank

178

re-ranked

Family-slot routing

proximity / vertical: re-ranked
support/contact: source order

Joint evaluation

Exact-label Recall@K

Verifier-derived Violation@K